

Evaluating the Effect of the Color-Word Stroop Test and VR as a Psychological Stressor

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ABSTRACT

Virtual Reality (VR) makes use of psychological stressors to enable users to feel immersed in different domains. Although stressor tests, like the Color-Word Stroop Test (CWST), have been shown to be valid and reliable, there are limited deployments of such stressors that exhibit appropriate immersion and user experience in VR. In this paper, we present the development and evaluation of two VR simulators of the CWST. We conducted a between-subjects study with 27 participants who completed the conventional CWST, a VR-only simulator of the CWST, and a gamified VR simulator of the CWST. We measured and conducted a statistical analysis of participants' CWST scores, perceived stress, immersion, user experience, playability, and heart rate. Our results show that there are no significant differences in using the two CWST simulators of the VR. We found that users' heart rate is significantly higher when using the VR simulators than the conventional CWST and that the VR game simulator significantly shows a higher score in immersion than the VR-only simulator. However, users' perceived stress was significantly less when using the VR game simulator. Finally, we reflect on design insights for stressors tests in VR and discuss directions for future work.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Stress; Virtual Reality; Stroop Color; Sports

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1 INTRODUCTION

VR research shows that immersion is important for specific contexts [20] and that existing VR video games have been successfully used to transfer real-life situations to training contexts [3, 29]. Although VR aims at providing a realistic experience, immersion in a virtual environment has other benefits such as spatial understanding. As opposed to other technologies, VR provides depth cues [2], and reduces the information clutter, typical in mobile or desktop applications. Nonetheless, virtual worlds can also become easily cluttered by adding too much information.

Multiple tests have been presented in the literature to generate physiological stress reactions in users for research purposes. These tests are called stressors. Some researchers divide stressors into two types: physical and psychological [4]. Physical stressors are tests that represent a challenge or change to the user's metabolism (e.g., exercise, exposure to cold or heat) and are not very well accepted, because can hurt the subjects. In contrast, psychological stressors are those that are not necessarily a challenge to the metabolism, but consider a cognitive load for the user (e.g., math tests, reverse spelling).

For this reason, multiple psychometric instruments have been developed to be psychological stressors such as the Trier Social Stress Test (TSST) [15], Serial Sevens test [26], or a Reverse Spelling test [18]. These tests aim at enabling the users to identify their internal state and be able to modify it. One of the most widely used instruments in the cognitive assessment of the executive aspects of attention control is the Color-Word Stroop Test (CWST) [28]. CSWT is a neuropsychological task used to assess the ability to inhibit cognitive interference, which occurs when the processing of a specific stimulus impedes the simultaneous processing of a second stimulus. This effect is known as the *Stroop Effect* [25]. In the CSWT, the participant has to indicate the color of the word (e.g., Blue), while ignoring the semantic content of the word (e.g., Red). The CWST instrument has been used as a tool for stress-related research ever since and is considered one of the most widely used methods to generate cognitive stress [1].

With the advent of wearable sensors, physiological stress reactions can be increasingly monitored [8]. The heart rate (HR) has

been reported as the most prominent signal to adequately determine stress reactions [14]. Although it has been shown that the realistic experience provided by VR enables the replication of real life-situations and application development such as training and therapy for phobias [9], to the best of our knowledge experiences using immersion techniques that can transfer psychological stressors are currently scarce [8].

In this paper, we hypothesize that CWST using VR can be used as an alternative to the conventional test, and can be used as a psychological stressor. In this work, we present the design and evaluation of two CWSTs using VR, one of them using the conventional approach, and the second one using a video game approach. The contributions of this paper are:

- Two open-source VR simulators that implement the Stroop color test for use in laboratory testing.
- A set of immersive approach that transfer the the conventional Stroop color test to a real scenario and can help other researchers implement them in their scenarios.
- Empirical quantitative evidence that our proposal is a valid test and comparable to the conventional Stroop color test.

2 RELATED WORK

2.1 Psychological Stressors used in VR

Research has shown that the use of VR is useful for providing realistic experiences, and it can be helpful in the replication of real-life situations to obtain different physiological stress responses [16, 19, 27, 31]. Martens et. al. [17] studied the stress reactions with the implementation of two VR elevators (external and control). In the external elevator, they simulated going up on the outer side of a tall building, and in the control, participants were going up inside the building. The authors compared the physiological responses between both VR elevators. The results of an evaluation with 28 healthy men showed that the external elevator increased skin conductance, and pulse, altered heart rate variability, and a delayed rise in cortisol.

Other research projects have used different psychometric tests as physiological stressors [13, 30]. Fallon et al. [5] evaluated the physiological stress responses between VR, In-vivo, and conventional versions of the TSST. Their findings show a slower heart rate and heart rate variability response in VR TSST compared to the in-vivo TSST. This body of work shows that while it is possible to apply psychological stressors using VR, some of them involve a very complex procedure and are difficult to apply. However, there are others such as the CWST that have been used a lot to generate these same physiological reactions and its application is simpler and easier to replicate in a controlled environment.

2.2 Color-Word Stroop test and VR

The application of the CWST has been implemented in multiple computational systems using VR in order to activate the user before or after a condition to be tested [10, 22]. Rizzo et al. [24] developed "The Virtual Classroom", a VR environment to compare performance data for assessment, treatment, and rehabilitation. They designed a CWST for the purpose of assessing users' complex attention. They demonstrated that a VR CWST can have the same interference effect as a conventional CWST.

On the other hand, Gradl et al. [7] designed the *StroopRoom*, a hexagonal room based on the VR CWST using virtual world elements. Their results show that a VR environment is able to implement the CWST by increasing physiological characteristics such as heart rate and skin conductance. However, usability, user experience, and acceptance factors of VR were not evaluated.

These works illustrate that the CWST could be used in VR environments. However, there are still open questions about the mechanics used in the design and implementation of this test, as well as the effect on the subjective experience of the users in this new modality.

3 VR STROOP TEST

In this section, we present two versions of the CWST test that can be implemented in a VR environment using a football scenario.

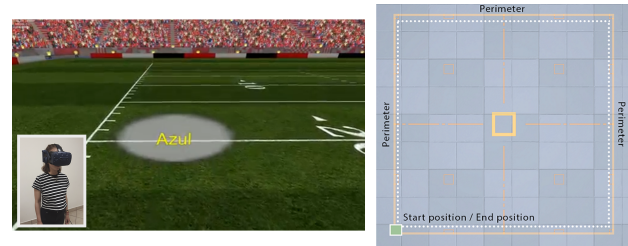


Figure 1: Virtual Stroop test scenario; the left figure shows the player's view, in front of which is shown the word to be mentioned (Stroop test). The right figure shows the configured perimeter (route) to follow.

3.1 Simulator 1: Virtual Stroop test

The virtual Stroop test is a 3D scenario that places the player in a white reality environment. In front of her there is a text box with the word "Ready?".

Once the instructor starts the test, the player begins to move over a perimeter (in the virtual environment) predefined by the therapist. We define the perimeter of a box 360 ft. wide by 160ft. long to a scale of 2:1 (Figure. 1). After that, the player starts moving at a predefined speed per minute.

While the player is moving, the box displayed in front of her starts emulating the Stroop test. That is, it displays a written word describing color but colored in a different color than the one it is written in, then, the user must say the word that is displayed in front of her.

3.2 Simulator 2: Virtual Stroop Game

Stroop Game is a 3D scenario that places the player at the center of a virtual white environment. In front of her there is a table with a ball of 4 inches in diameter (on the virtual scale). Behind the table, they are presented with three text boxes, one box on top and two boxes at the bottom.

When the player starts the Stroop Game, they begin with a training phase. The training consists of a series of instructions given by the computer which mention the following:

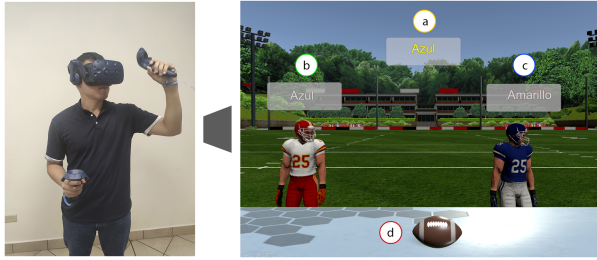


Figure 2: View of the player inside the simulator 2 (Stroop Game).

- The text box located at the top (Figure. 2.a) will display the name of a color (e.g., blue) but in a different font color (e.g., red).
- The text boxes at the bottom, i.e., bottom left box (Figure. 2.b) and bottom right box (Figure. 2.c) will show one word each. One box will present the word written in the top box and the other will show the color of the top word.
- The player must select one of the bottom boxes that corresponds to the color of the word at the top (this is considered the correct square), ignoring the other word.
- To select the box, the user should throw the ball shown above to the correct box (Figure. 2.d).

4 METHODS

To evaluate whether both Stroop-Color VR proposals can be used as alternatives of conventional Stroop-Color test, we conducted a between-subjects study using the following hypotheses:

- H1:** The Stroop-color score will not be significantly different when using the CWST VR proposals than when doing it in the conventional way.
- H2:** The heart-rate data measures will not be significantly different when using the CWST VR proposals than when doing it in a conventional way.
- H3:** The user experience and playability scores will not be significantly different when using the CWST VR proposals than when doing it in a conventional way.
- H4:** The perceived stress scores will not be significantly different when using the CWST VR proposals than when doing it in a conventional way.

4.1 Participants

We recruited 27 adults (women = 13; mean age = 26 years old, SD = 3.61) for the study. Our participants' ages were between 18 and 35 years old. Our exclusion criteria were all those adults who had heart conditions or were under medication for stimulants or autonomic nervous system depressants. We asked our participants not to ingest caffeine at least six hours prior to the study.

4.2 Experimental setup

Both simulators are implemented in Unity3D (Unity Technologies, San Francisco, USA), the default environment is designed with basic Unity elements to make it lightweight and to use few computer

resources. We use HTC VivePro VR headset with two positioning sensors placed 8 ft. high with a 45-degree downward tilt. Before starting the tests, we calibrated the height and positioning of the headset with VivePro software (HTV's proprietary VivePro). To use the headset inside Unity 3D, we used the SteamVR plug-in (version 2.7.3). To interact with the environment we use the two controls included with the VR headset.

4.3 Procedure

Our participants were randomly allocated to carry out one Stroop-color test under one of these conditions (a) 9 participants doing the Stroop-color test with the simulator one using a VR environment (i.e., vr-simple group); (b) 9 participants using the simulator two (i.e., vr-game group); and (c) 9 participants doing it in a conventional way without a VR environment (i.e., control group).

Each session lasted 16 minutes. This total time was divided into five blocks. The first block lasted 5 min and aimed to inform participants about what they were going to do during the session, ask them for authorization to take photos and videos during the test, and to place a H10 band for heart rate data measurement. The second and fourth blocks lasted 3 min and involved a rest period where the participants watched a relaxing video. On the other hand, the third block lasted on average 3 min, and each participant took the Stroop-color test corresponding to the assigned condition. To finish the session, block five had a duration of approximately 2 min, and consisted of performing two reading tasks to measure the normal diction of each participant.

4.4 Data collection

We used the following instruments for calculating the Stroop-color score, heart-rate measures, user experience and playability and perceived stress of our participants:

Stroop-Color score. To evaluate the Stroop-color score in the conventional test and virtual Stroop test, we count the correct words mentioned manually. On the contrary, for the Stroop game test, the system allows us to calculate them automatically, as well as the number of hits¹. For the three conditions, we corroborated the results with video recordings from each session. Finally, we calculated the R-int value. The R-int value is a score obtained from the difference between the Word-Color condition score (PC score) and the performance score in the Word and Color conditions [6].

Heart-rate measures. We used a Polar's H10 band with the Bluetooth protocol as a physiological sensor to collect the user's heart rate data. To reduce noise and atypical values including misreadings with empty values, we extracted and processed the heart rate data for each activity phase (9 minutes in total).

User experience and playability. To assess the participants' experience using the different Stroop-Color modalities, we used three different instruments. First, we apply the main module of the game experience questionnaire (GEQ) [11]. GEQ's scores are calculated as the mean value of its items. Each item uses a 5-point scale (0 = Not at all; 4 = Extremely). The second instrument applied was the user experience questionnaire (UEQ) [23]. This questionnaire uses bipolar terms (e.g., annoying - enjoyable) scored on a 7-point scale, from -3 to 3, with 0 being a neutral response.

¹It is considered a hit if the user selects the color with the ball and not the word

Finally, to assess immersion feeling, we used the immersive experience questionnaire (IEQ) [12]. The IEQ uses a 5-point scale (Nothing = 1; Completely = 5). The overall score ranges from 31 (worst) to 155 (best).

Perceived stress. To evaluate the perceived level of stress provided by the Stroop-color modalities we used the stress scale of the DASS-21 questionnaire before and after each session [21]. This stress scale consists of 7 items. Each item uses a 4-point scale (0 = It has not happened; 3 = it has happened a lot, or most of the time).

4.5 Data Analysis

We followed a quantitative approach to test the significance of our results. We used the Shapiro-Wilk Normality Test to confirm that all our data have a normal distribution. For our data on Stroop score, heart rate, user experience, and playability we used an ANOVA test. Furthermore, for the perceived stress analysis we used a Two-way mixed ANOVA. Both Statistical tests were accompanied by a posthoc Tukey correction.

5 RESULTS

H1: Stroop Score. The results obtained show that participants scored similarly across all conditions when calculating the R-int value: conventional ($mean = -3.1925$, $SD = 21.029$), Virtual Stroop test ($mean = -2.9933$, $SD = 25.582$), Virtual Stroop game ($mean = -2.8804$, $SD = 14.167$). We found no significant differences in the score obtained in the tests (f -value = 0.00986, $df = 2$, p -value = 0.9901).

H2: Heart Rate. Our results show that all conditions increased the heart rate of the participants; conventional ($mean = 84.87$, $SD = 10.8$), virtual - simple ($mean = 89.42$, $SD = 12.292$), virtual - game ($mean = 103.13$, $SD = 15.354$). We found a significant difference in the conditions in the average heart rate (f -value = 4.84434, $df = 2$, p -value = 0.01). When we performed the Post Hoc Tukey correction, we found that the difference was between the conventional condition and the virtual-game condition (p -value = 0.01).

H3: User Experience and Playability. According to the UEQ, our participants of both VR conditions scored similar user experiences compared to participants of the conventional condition. Thus, an ANOVA statistical test shows no significant difference in the mean scores of attractiveness (f -value = 0.199, $df = 1$, p -value = 0.659), stimulation (f -value = 2.12, $df = 1$, p -value = 0.158), perceptivity (f -value = 0.003, $df = 1$, p -value = 0.96), efficiency (f -value = 0.221, $df = 1$, p -value = 0.642), dependability (f -value = 0.034, $df = 1$, p -value = 0.855) and novelty (f -value = 1.671, $df = 1$, p -value = 0.208).

On the other hand, our results from the core gaming module of the GEQ questionnaire, show that all the Stroop-Color tests have a positive and enjoyable gaming experience. We found that the Virtual Stroop Game condition scored slightly higher ($mean = 2.0994$, $SD = 0.2588$) than the Virtual Stroop test ($mean = 1.9224$, $SD = 0.5123$) and the conventional conditions ($mean = 2.0433$, $SD = 0.2407$). An ANOVA statistical test shows no significant difference between the three conditions (f -value = 0.11, $df = 1$, p -value = 0.0743). Moreover, according to the IEQ questionnaire, participants of the Virtual Stroop game condition scored a higher

immersion feeling ($mean = 113$, $SD = 10.75$) when compared to the participants of the Virtual Stroop test condition ($mean = 93.55$, $SD = 14.31$). A T-test for independent measures shows a significant difference between both virtual conditions ($t = -3.2582$, $df = 14.854$, p -value = 0.005351).

H4: Perceived Stress. The results DASS-21 questionnaire show that our participants reported their perceived stress similarly previous to and after the session. A Two-way mixed ANOVA shows no significant difference between conditions (f -value = 2.658, $df = 2$, p -value = 0.08) and, the pre and post-measures of each condition (f -value = 5.35, $df = 1$, p -value = 0.564).

Correlations. We computed the Pearson's correlation coefficient and found a positive correlation between the Heart-Rate measures and the immersion score, where the higher the immersion, the higher the Heart-Rate ($r = 0.3915$, p -value = 0.04341).

Also, we computed the correlation coefficient between the immersion score and the (post-experiment) perceived stress. However, it was not significant ($r = -0.4592$, p -value = 0.0552). This could be because the time to the stimuli exposure was too short to cause an effect on the perceived stress.

6 DISCUSSION

Overall, the evaluation results indicate that all our participants scored similarly on the Stroop test, regardless of the condition in which they participated (conventional, Virtual Stroop test, Virtual Stroop game). This means that the 3 tests are comparable in terms of the scores obtained. This corroborates our hypothesis one (H1) and means that the virtual simulators developed can be an application option in terms of the traditional test. We also identified that the participant's heart rate increased in all conditions. However, participants were more activated using the Virtual Stroop game when compared to the Virtual Stroop test. Although this rejects our hypothesis two (H2), this difference is positive if the goal is to activate the user in a non-aggressive manner.

Finally, we identified that participants gave similar scores on user experience, playability, and perceived stress which corroborates hypotheses three (H3) and four (H4). Since participants perceive the three Stroop test options as enjoyable, fun, and with a good user experience. In contrast to related work, our simulator proposal represents a new game mechanic. This mechanic allows the CWST test to be incorporated into new contexts (offices, school classrooms, sports centers) without limiting the environment under investigation. We believe that our two simulators can be an alternative to the CWST test if the intention is to use a test that is similar to the traditional test and activates users in a non-negative way.

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